RDF4RiskAssessment Toolkit: A toolkit for converting tabular research data to FAIR RDF for risk assessment and life sciences

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Abstract

Risk assessment is crucial for consumer health protection and food safety within the Risk Analysis framework. It requires integrating heterogeneous data from multiple scientific disciplines. However, research data often exists in non-standardized formats, creating data silos that hinder cross-institutional reuse and AI-based analysis. We developed the RDF4RiskAssessment Toolkit within the KIDA project to transform tabular research data into FAIR-compliant Linked Data and used it in the AMBROSIA project to harmonize concepts. The toolkit features a four-stage workflow: matching table generation, term reconciliation via external ontology services, configurable RDF generation, and bidirectional RDF-table conversion. It supports metadata standards (DCAT, Dublin Core, BIBO) and vocabulary alignment (SKOS). Testing with 14 datasets from 6 publications yielded approximately 1,860 mapped vocabulary concepts. The toolkit enables researchers without Semantic Web expertise to produce machine-readable, interoperable datasets for risk assessment and life sciences.

Keywords

Risk Assessment, Life Sciences, Consumer health protection, Knowledge Graph, Linked Data

1. Introduction and Tool Description

Risk assessment is central to consumer protection and food safety within the Risk Analysis framework. It increasingly depends on the integration of data from chemistry, life sciences, environmental sciences, and social sciences. Despite the growing availability of data, research often relies on spreadsheets or isolated databases, leading to data silos and reduced interoperability. This fragmentation hinders AI-based methods, collaborative evaluation, and effective decision-making. Several initiatives reflect the need for improved data interoperability. One Health promotes the integration of human, animal, and environmental health data, whereas the Adverse Outcome Pathway (AOP) framework provides a structured toxicological representation [1, 2]. The FAIR principles (Findability, Accessibility, Interoperability, Reusability) and technologies such as Knowledge Graphs and Linked Data offer solutions through ontologies and standards. Existing tools such as OpenRefine, Karma, and Eccenca support Linked Data transformation but assume stable data pipelines and require expert knowledge. These tools often rely on semi-automated string matching, which works well with broad vocabularies such as Wikidata but struggles with domain-specific ontologies. Furthermore, schema-driven approaches (RML, YARRRML) are difficult to adapt to the variable structures typical of interdisciplinary research. They are not well-suited for the small-scale, heterogeneous datasets common in toxicology, One Health, and food safety research.

To address this gap, we developed the RDF4RiskAssessment Toolkit within the KIDA project, a pilot project aimed at accelerating AI and data interoperability adoption among German Federal research institutes. Additionally, the toolkit was used to harmonize concepts from agriculture and predictive

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microbiology in the Ambrosia Project. The toolkit converts tabular data into machine-readable Linked Data through user-friendly workflows that do not require prior Semantic Web expertise.

This was achieved through a web-based interface comprising four transformation services. The first service is the "Matching Table Generation" service. Users upload CSV/Excel files, and the application generates a semi-automatically structured view of headers and values, producing a cleaned dataset and SSSOM-like candidate term table. Within the Reconciliation Service, the candidate term table is used to link the dataset terms to authoritative ontologies via API endpoints (Wikidata, BioPortal, AgroPortal, EarthPortal, NCBI). Users can select endpoints and term matches in a semi-automatic interface that supports string matching via similarity metrics (Levenshtein). Once matching is complete, the original data and reconciled candidate term table are used by the RDF Generation service. The content of both tables is combined, and a table-shaped RDF, where rows become subjects, columns become predicates, and cells become objects, enriched with reconciliation data is generated in TriG, Turtle, or JSON-LD formats. The mapped terms are stored automatically in a SKOS representation to preserve and harmonize the matchings across datasets. For each converted RDF dataset, DCAT Metadata is generated. Optionally, a custom mapping schema can be used for advanced use cases supporting blank nodes and nested structures. Finally, we provide the RDF-to-Table Conversion service, which allows us to reverse the transformation, enabling validation and accessibility for users without SPARQL expertise by reconstructing nested resources in a tabular format.

By enabling semantic enrichment and improving cross-disciplinary data interoperability, the RDF4RiskAssessment The toolkit contributes to more accessible, reusable, and integrated scientific datasets for risk assessment and life sciences. Throughout the KIDA project, the toolkit supported numerous service requests by converting the research data into Linked Data.

2. Results

Within the KIDA project, 14 datasets from 6 publications were converted into Linked Data, yielding approximately 1,860 vocabulary concepts mapped against Wikidata and OBO ontologies. The largest data graph contained 140,699 triples; the smallest contained 189. Data spanned diverse domains from wine authenticity to epidemiological analyses, demonstrating the toolkit's adaptability across domains and scales. The generated Knowledge Graph supports SPARQL queries, cross-disciplinary data exchange, and the storage of relevant metadata and publication links. In AMBROSIA, the initial SKOS vocabularies were reconciled against external ontologies using the Reconciliation Service, enabling ongoing term enrichment and mapping (e.g., Wikidata, OBO) Source code is available at: https://github.com/KIDA-BfR/RDF4Risk-Toolkit

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